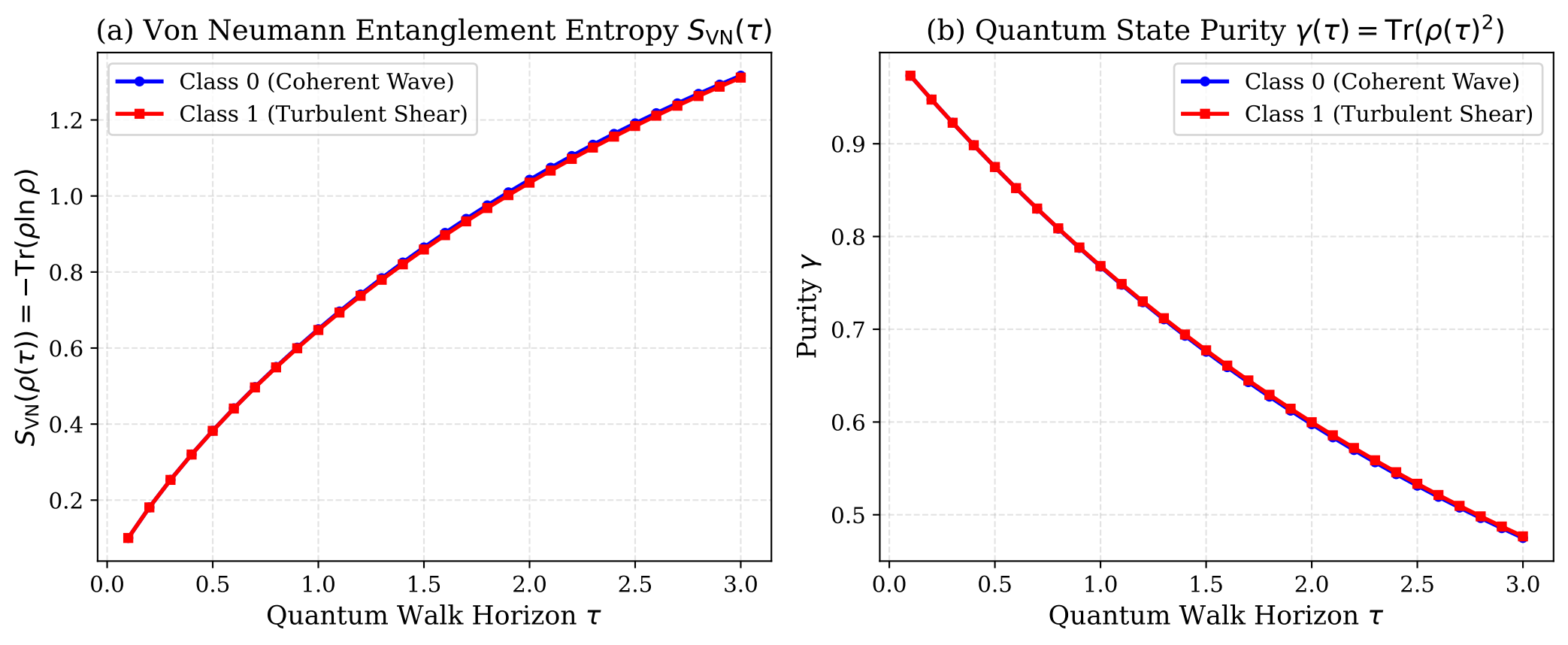
(a) Von Neumann Entanglement Entropy $S_{\mathrm{VN}}(\tau)$
Class 0 (Coherent Wave)
Class 1 (Turbulent Shear)
$S_{\mathrm{VN}}(\rho(\tau)) = -\mathrm{Tr}(\rho \ln \rho)$
Quantum Walk Horizon τ
(b) Quantum State Purity $\gamma(\tau) = \mathrm{Tr}(\rho(\tau)^2)$
Class 0 (Coherent Wave)
Class 1 (Turbulent Shear)
Purity γ
Quantum Walk Horizon τ